# Checkpoint 5 · Prototype Ready & User Flows

System Architecture, Screen Mockups, Storyboard & Test Plan | INT335 Design Thinking

## 1. System Architecture & Prototype Overview

ReturnLoop LPU is structured as a responsive full-stack progressive web application engineered specifically for mobile devices with ■0 infrastructure dependencies. It enables rapid lost/found logging, real-time trait matching, and campus security admin verification.

**Figure 1: 6-Panel User Storyboard**

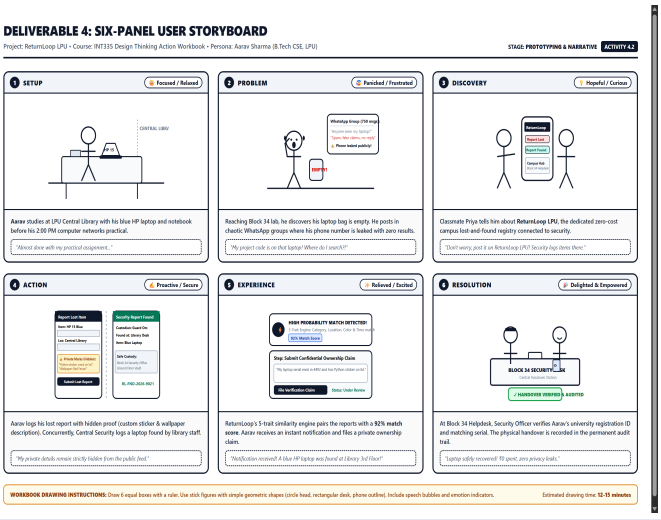



**Figure 2: Architecture & User Flow**

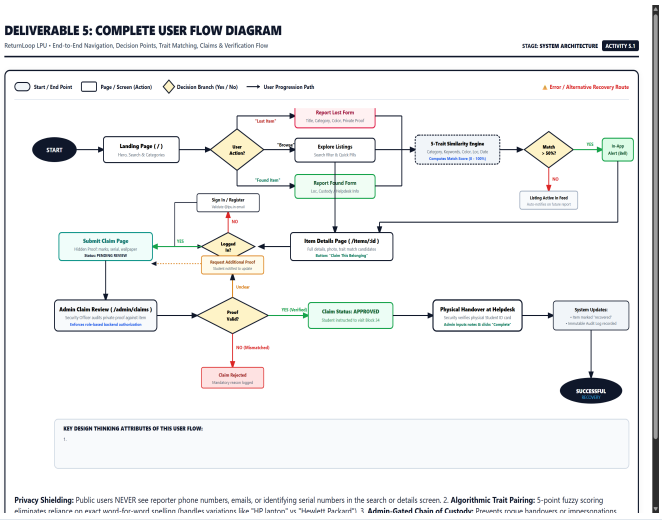
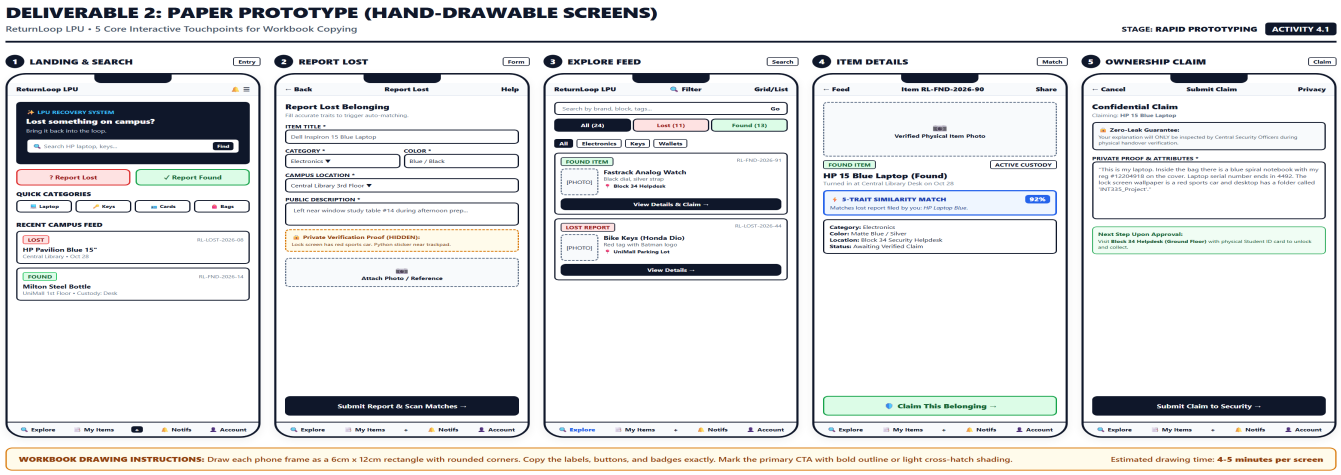



**Figure 3: Paper Prototype Screens (Section 13 Compliance)**



## 2. Primary User Scenarios & Prototype Verification Protocol

| Scenario # | User Goal & Persona | Step-by-Step Prototype Flow | Success Criteria |
|---|---|---|---|
| **Scenario A** | **Report Lost Calculator**<br>Aarav (Hosteller) | 1. Tap '+ Report' on mobile navbar.<br>2. Select 'Lost Belonging'.<br>3. Choose 'Electronics' & input model.<br>4. Enter secret identifier (scratched initials). | Report completed in < 60s; secret proof hidden from public view. |
| **Scenario B** | **Report Found Keys**<br>Priya (Day Scholar) | 1. Open quick report modal.<br>2. Select 'Found Item' & Block 34.<br>3. Upload photo & set drop status.<br>4. Receive automated match notification. | Instant notification triggered; no personal phone number exposed. |
| **Scenario C** | **Admin Handover**<br>Guard Gurpreet | 1. Open Admin Desk portal.<br>2. Compare owner's submitted proof against physical item.<br>3. Click 'Verify & Handover'. | Audit trail logged; 0% chance of handover to impostors. |